

Measuring Learning in Informative Processes

Robert C. Luskin, Ariel Helfer, and Gaurav Sood

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Abstract

We often want to know how much people learn from informative processes like election campaigns and deliberative forums, and who learns the most. The obvious measure, the gain in the share of knowledge items answered correctly, can be a poor indicator of true learning. We show, in a linear model, a nonlinear one and a general one, that when learning is nondecreasing in initial knowledge and an independent learning shock, post-process knowledge has nonnegative covariance with true learning, overall and conditional on true initial knowledge. Observed gain has no such unconditional guarantee and can be negatively correlated with learning. Conditioning on noisy observed initial knowledge does not inherit the guarantee. In 1,212 simulated processes whose observable features match 21 Deliberative Polls, observed gain’s median correlation with true learning is .01 and post-process knowledge’s is .49. The choice matters: in the same polls, the above-median education difference in observed gain is uncertain (.017 [-0.005, .041]), while their post-process knowledge is higher conditional on initial knowledge (.066 [.049, .085]). These comparisons do not identify the amount learned.

1 Introduction

Many communication processes are broadly informative, exposing their participants to some goodly quantity and wide variety of information on at least one sizable topic, like policy or electoral choices. The category includes both natural processes like election campaigns and ad hoc but naturalistic ones like Deliberative Polls. The information exposure need not be the point. Often, it is not. In election campaigns, it is incidental, a by-product of persuasion or mobilization. In Deliberative Polls and other deliberative fora, it is intrinsic but instrumental—not an end in itself but an ingredient in more fully considered, better founded opinions. But in whatever fashion, for whatever reason, the exposure must occur, and at least some participants must learn something from it, or the process would be only degenerately “informative.”

Almost always, we are interested in this learning—sometimes in the learning of this or that particular bit, affecting this or that particular attitude (Gilens 2001), but more often in the amount of topic-wide learning, affecting a wider range of attitudes, as well as vote intentions, levels of engagement, and the like (Bartels 1996; Delli Carpini and Keeter 1996; Althaus 2003). In the aggregate, this is also normatively important, speaking to the educative value of election campaigns and the deliberative quality of deliberative fora.

But how best to measure it? The obvious choice is observed knowledge gain: the increase in the proportion of knowledge items answered correctly. For aggregate description, *faute de mieux*, this may serve. But individual-level analyses—of learning as a function of causally prior variables or of causally subsequent ones as a function of learning—are another story. There, observed knowledge gain turns out to be a surprisingly weak indicator of true knowledge gain. This paper shows

why in two models, a linear one and a nonlinear one, then generalizes the key results to any measurement function, gives conditions for post-process knowledge to track learning and compares the proxies under specified learning models, measures how much better in simulations calibrated to real Deliberative Polls, and shows that the choice changes substantive answers.

2 Observed and True Knowledge

The knowledge we are used to thinking about is “observed.” It is, by definition, all we ever see. Conventionally, the i th person’s observed knowledge at time t —call it x_{it} —is the proportion of the questionnaire’s knowledge items the person answers correctly. We are much less used to thinking about “true knowledge”—what observed knowledge is trying to observe. Its depths can be left to philosophy and neuroscience. Nearer the surface, we may define the i th person’s true knowledge at time t —call it X_{it} —as the proportion of all knowable, relevant items he or she knows at time t . Observed knowledge gain is $\Delta x_i = x_{i2} - x_{i1}$ and true knowledge gain $\Delta X_i = X_{i2} - X_{i1}$.

The set of knowable, relevant items includes everything that is relevant and knowable, whether any mortal knows it or not; on topics like policy issues, it must number at least in the hundreds of thousands. So X_{it} never gets much above 0, and since very few people follow politics very closely, its distribution must be severely right-skewed. By contrast, x_{it} typically ranges all the way from 0 to 1, because the questionnaire’s items are a very small, very easy subset of what is knowable. Question writers do not know enough to ask the items known only by policy experts, let alone those no one knows. Of those they know, they favor relatively easy ones; otherwise, hardly anyone would score much above 0, except by lucky guessing.

These facts suggest the following properties of knowledge and learning:

- L1. The participants learn on average: $E(X_{i2}|X_{i1}) > X_{i1}$ for $X_{i1} > 0$.
- L2. The more knowledgeable learn more: $E(\Delta X_i|X_{i1})$ increases with X_{i1} . More new information sticks when there is more previously cognized information for it to stick to (Recht and Leslie 1988; Eckhardt et al. 1991; Cooke et al. 1993; Hambrick 2003).
- L3. Those who know nothing learn nothing.
- M1. Those who know nothing show no knowledge, and those who know the most show as much as the measure permits.
- M2. The more knowledgeable appear to know more, but the less so, the more they know: $E(x_{it}|X_{it})$ increases with X_{it} but flattens as it approaches its ceiling of 1.
- M3. People appear to know more than they do: $E(x_{it}|X_{it}) > X_{it}$.
- M4. The more knowledgeable appear to learn more, then, beyond a point, to learn less. Those who know nothing actually learn nothing; those who know the most, despite actually learning the most, cannot show it, having started at or near the ceiling.

Three phenomena are jointly responsible for the results that follow.

Item-sampling bias. A questionnaire’s knowledge items are hardly a random draw from the universe of relevant, knowable items. They are much, much, much easier than average. Question writers cannot even think of the items known only by policy experts, much less those nobody knows. Among those they can think of, they tend to choose items that anyone paying a modicum of attention to the news should know. Otherwise, hardly anyone would score much above 0, except by lucky guessing. Thus many of the relatively knowledgeable may ace a questionnaire’s knowledge items, letting x_{it} range far higher than X_{it} , usually all the way up to 1.

Ceiling effects. The participants who already know a lot, having a relatively high X_{i1} and thus an absolutely high x_{i1} , cannot show much observed knowledge gain. For true knowledge, there is always room for improvement; for observed knowledge, not so.

The rich get richer. Knowing more about something makes it easier to learn still more (Dooling and Lachman 1971; Bransford and Johnson 1972; Recht and Leslie 1988; Eckhardt et al. 1991; Cooke et al. 1993; Hambrick 2003). Learning is easier when the material falls effortlessly into context. Hence the participants who cannot show much observed knowledge gain are precisely those whose true knowledge gain should be greatest.

These considerations yield two second-blush intuitions. First, ceiling effects and the rich getting richer should depress the correlation between Δx_i and ΔX_i . Second, x_{i2} may be a better indicator. Those who emerge observably knowing a lot have generally gained a lot—either observably, if they initially knew only a little, or unobservably, if they initially knew a lot.

3 A Linear Model

Let

$$x_{i1} = a_1 + b_1 X_{i1} + u_{i1}, \quad (1)$$

$$x_{i2} = a_2 + b_2 X_{i2} + u_{i2}, \quad (2)$$

$$X_{i2} = \alpha + \beta X_{i1} + v_i, \quad (3)$$

implying

$$\Delta x_i = (a_2 - a_1) + b_2 \alpha + (b_2 \beta - b_1) X_{i1} + (u_{i2} - u_{i1}) + b_2 v_i, \quad (4)$$

$$\Delta X_i = \alpha + (\beta - 1) X_{i1} + v_i. \quad (5)$$

For the conditional-mean interpretations, assume $E(v_i | X_{i1}) = 0$ and $E(u_{it} | X_{i1}, v_i) = 0$, with u_{i1} and u_{i2} uncorrelated. These assumptions imply the zero covariances used below; zero covariances alone would suffice for the covariance calculations. The variances $\sigma_{X_1}^2$, $\sigma_{u_1}^2$, $\sigma_{u_2}^2$ and σ_v^2 are all strictly positive: there is some measurement error at each wave, true initial knowledge does not perfectly predict true later knowledge, and true initial knowledge varies. The linear equations are local approximations, not a globally bounded response model.

A second layer of assumptions expresses the three phenomena. *Item-sampling bias.* Those who know nothing, and do not guess, score 0, so $a_t = 0$ and $b_t = E(x_{it} | X_{it} = X_{Mt}) / X_{Mt}$, where X_{Mt} is the highest true knowledge in the population. Since the questionnaire lets $E(x_{it} | X_{it} = X_{Mt})$ approach 1 while X_{Mt} is near 0,

$$b_1, b_2 > 1. \quad (6)$$

The rich get richer. A given difference in true initial knowledge results in a bigger difference in true later knowledge:

$$\beta > 1. \quad (7)$$

Ceiling effects. The more one knows, the less one can appear to gain: X_{i1} 's coefficient in (4) is negative,

$$b_1 > b_2 \beta. \quad (8)$$

The correlations of observed gain and of post-process knowledge with true gain are $\rho_{\Delta x \Delta X} = \sigma_{\Delta x \Delta X} / (\sigma_{\Delta x} \sigma_{\Delta X})$ and $\rho_{x_2 \Delta X} = \sigma_{x_2 \Delta X} / (\sigma_{x_2} \sigma_{\Delta X})$, where single-subscripted, squared σ 's are variances and double-subscripted ones covariances. From (1)–(5),

$$\sigma_{\Delta x \Delta X} = (\beta - 1)(b_2\beta - b_1)\sigma_{X_1}^2 + b_2\sigma_v^2, \quad (9)$$

$$\sigma_{x_2 \Delta X} = (\beta - 1)b_2\beta\sigma_{X_1}^2 + b_2\sigma_v^2, \quad (10)$$

$$\sigma_{\Delta x}^2 = (b_2\beta - b_1)^2\sigma_{X_1}^2 + \sigma_{u_1}^2 + \sigma_{u_2}^2 + b_2^2\sigma_v^2, \quad (11)$$

$$\sigma_{x_2}^2 = b_2^2\beta^2\sigma_{X_1}^2 + b_2^2\sigma_v^2 + \sigma_{u_2}^2. \quad (12)$$

Three results follow (proofs in Appendix A).

Result 1. $\rho_{\Delta x \Delta X}$ can be negative. Specifically,

- Condition 1a: $\rho_{\Delta x \Delta X} < 0$ if and only if $(\beta - 1)(b_1 - b_2\beta)\sigma_{X_1}^2 - b_2\sigma_v^2 > 0$.

Result 2. $\rho_{x_2 \Delta X}$ is always positive.

Result 3. $\rho_{\Delta x \Delta X}$ can be less than $\rho_{x_2 \Delta X}$. Each of the following is sufficient:

- Condition 3a: $(b_1 - 2b_2\beta)\sigma_{X_1}^2 + \sigma_{u_1}^2 > 0$;
- Condition 3b: $b_1 \geq 2b_2\beta$;
- Condition 3c: Condition 1a;
- Condition 3d: $\gamma_{21} < .5$, where $\gamma_{21} = \sigma_{x_1 x_2} / \sigma_{x_1}^2$ is the slope of the population regression of x_{i2} on x_{i1} .

Three points need noting. First, $\rho_{\Delta x \Delta X}$ need not be negative to be depressed: as the parameters approach a configuration under which it is negative, observed gain is enfeebled, positively but weakly correlated with true gain. Second, none of Conditions 3a–3d is necessary. Third, Condition 3b implies 3a, but Conditions 3a and 3c do not imply one another; neither is uniformly easier to satisfy. Condition 3d is directly checkable: in every one of the 21 Deliberative Polls we examine below, the observed correlation of x_{i1} and x_{i2} is between .11 and .72, and the regression slope of x_{i2} on x_{i1} is below .5 in 10 of them.

Guessing. Let people who do not know an item answer it correctly with probability κr_i , where r_i is the person's propensity to guess, the same at both waves and independent of the other variables, and κ the chance of a lucky guess. Observed knowledge becomes $y_{it} = x_{it}(1 - \kappa r_i) + \kappa r_i$. Then $\sigma_{\Delta y \Delta X} = (1 - \kappa E(r_i))\sigma_{\Delta x \Delta X}$ and $\sigma_{y_2 \Delta X} = (1 - \kappa E(r_i))\sigma_{x_2 \Delta X}$: guessing scales both covariances by the same nonnegative factor. Results 1 and 2 retain their strict signs when the factor is positive; guessing correctly with probability one makes the observed score constant.

4 A Nonlinear Model

The knowledge proportions x_{i1} and x_{i2} are bounded by 0 and 1 and often approach the ceiling; their difference lies between -1 and 1. Their consequently decreasing slopes on true knowledge (M2's ceiling) are a major part of the story, which the linear model captures only through $b_1 > b_2\beta$. A quadratic measurement equation makes the ceiling explicit. Let

$$X_{i2} = \beta X_{i1}, \quad (13)$$

$$x_{it} = bX_{it} + cX_{it}^2 + u_{it}, \quad t = 1, 2, \quad (14)$$

with $\beta > 1$, $b > 0$ and $c < 0$. Assume $E(u_{it} | X_{i1}) = 0$ and uncorrelated errors across waves, so that the errors are also uncorrelated with X_{i1}^2 . Then

$$\Delta X_i = (\beta - 1)X_{i1}, \quad (15)$$

$$\Delta x_i = (\beta - 1)bX_{i1} + (\beta^2 - 1)cX_{i1}^2 + u_{i2} - u_{i1}. \quad (16)$$

M2 requires observed knowledge to rise with true knowledge over the range of true knowledge at both waves: $b + 2cX_{it} > 0$, or $X_{M2} = \beta X_{M1} < -b/(2c)$; equality permits a zero derivative at the endpoint. $E(\Delta x_i | X_{i1})$, by contrast, peaks at $X_{i1} = -b/[2(\beta + 1)c]$, below X_{M1} only if the support extends beyond that point. M4 therefore requires $X_{M1} > -b/[2(\beta + 1)c]$ as well as $\beta > 1$. Writing V and C for variances and covariances,

$$C(\Delta x_i, \Delta X_i) = (\beta - 1)^2[bV(X_{i1}) + (\beta + 1)cC(X_{i1}^2, X_{i1})], \quad (17)$$

$$C(x_{i2}, \Delta X_i) = (\beta - 1)\beta[bV(X_{i1}) + c\beta C(X_{i1}^2, X_{i1})], \quad (18)$$

$$V(\Delta X_i) = (\beta - 1)^2V(X_{i1}), \quad (19)$$

$$V(\Delta x_i) = (\beta - 1)^2b^2V(X_{i1}) + (\beta^2 - 1)^2c^2V(X_{i1}^2) + 2bc(\beta - 1)(\beta^2 - 1)C(X_{i1}^2, X_{i1}) + V(u_{i1}) + V(u_{i2}), \quad (20)$$

$$V(x_{i2}) = b^2\beta^2V(X_{i1}) + c^2\beta^4V(X_{i1}^2) + 2bc\beta^3C(X_{i1}^2, X_{i1}) + V(u_{i2}). \quad (21)$$

Since $X_{i1} \geq 0$, $C(X_{i1}^2, X_{i1}) > 0$, and with $c < 0$ the second term in each bracket is negative. Let $\gamma = C(X_{i1}^2, X_{i1})/V(X_{i1})$, the slope of the population regression of X_{i1}^2 on X_{i1} . Then:

- R1. $\rho_{\Delta x \Delta X} > 0$ if and only if $\gamma < -b/[(\beta + 1)c]$.
- R2. $\rho_{x_2 \Delta X} > 0$ if and only if $\gamma < -b/(\beta c)$.
- R3. Since $-b/[(\beta + 1)c] < -b/(\beta c)$, $\rho_{\Delta x \Delta X} > 0$ implies $\rho_{x_2 \Delta X} > 0$, but not the reverse.
- R4. Under M2, R2's condition always holds: $\rho_{x_2 \Delta X} > 0$. The bracket in (18) is the covariance of X_{i1} with $bX_{i1} + c\beta X_{i1}^2$, a function that rises over the whole range of X_{i1} when $\beta X_{M1} \leq -b/2c$; the covariance of a variable with an increasing function of itself is positive. The bracket in (17) is the covariance of X_{i1} with $bX_{i1} + (\beta + 1)cX_{i1}^2$, which rises and then falls, and can be negative.

5 How General Are These Results?

A monotone item-response model extends the argument beyond the linear and quadratic approximations. Let each questionnaire item j have a difficulty d_j on the scale of true knowledge, and let a person know the item with probability $F(X_{it}, d_j)$, nondecreasing in X_{it} . In the simplest case F is a threshold—the person knows the item if and only if $X_{it} > d_j$ —and observed knowledge is $x_{it} = G(X_{it})$, where G is the distribution function of the questionnaire's difficulties. An easy questionnaire concentrates its difficulties low, so G rises quickly and flattens at 1. Evenly spread difficulties give a piecewise linear G ; a smooth, concentrated distribution gives something close to the quadratic; a logistic F gives an item response model. Let people who do not know an item answer it correctly by lucky guess. Item-response and guessing randomness are independent of initial knowledge and the learning shock, and guessing probabilities do not vary with either. Let true learning be $\Delta X_i = h(X_{i1}, R_i)$, where R_i is a scalar learning shock independent of X_{i1} , and h is nondecreasing in R_i .

Proposition 1. *If h is nondecreasing in X_{i1} (the rich get richer), $\text{Cov}(x_{i2}, \Delta X_i) \geq 0$.*

Proposition 2. *For any h , $\text{Cov}(x_{i2}, \Delta X_i | X_{i1}) \geq 0$.*

Proposition 1 strengthens L2: learning must be nondecreasing in initial knowledge at each value of the independent shock, not merely on average. Both propositions follow from the association of independent random variables: any two nondecreasing functions of independent variables have nonnegative covariance (Harris 1960; Esary et al. 1967). With finite second moments, neither argument requires a particular shape of G or distribution of true knowledge. Item-specific guessing probabilities are allowed under the stated independence assumptions (Appendix A). Observed gain has no such unconditional guarantee. Under deterministic measurement, $\Delta x_i = G(X_{i1} + \Delta X_i) - G(X_{i1})$ can rise and then fall in X_{i1} while true gain rises throughout. A ceiling alone does not require this shape over every possible support. With deterministic measurement, conditioning on X_{i1} fixes x_{i1} , so Δx_i and x_{i2} differ only by a constant. Conditioning on the realized observed baseline also makes gain and post-process scores equivalent, but does not condition on true initial knowledge. The two distinctions matter when baseline knowledge is measured with error (Lord 1967; Allison 1990).

6 How Large Are the Differences?

The propositions say nothing about magnitudes. To gauge them, we simulate processes and ask how well each proxy tracks true gain. Each simulated process has 300 participants, like a Deliberative Poll. The simulated true-knowledge variable is a right-skewed gamma trait, rather than a literal proportion of all knowable items, and item difficulties are drawn from a range of its quantiles, so that questionnaires run from very easy to moderately hard. People know items either by threshold or with logistic probability, and guess correctly with probability 0, .1 or .25. We vary the number of items (5 to 15), the median learning-rate multiplier and its log-scale dispersion, and three forms of learning: proportional to initial knowledge (L2), unrelated to it, and larger for those who know less.

Within the specified parameter ranges, we retain processes whose observable summaries resemble the polls. For each of 21 Deliberative Polls reported in Luskin et al. (2022), we compute five observable features: the mean and standard deviation of initial observed knowledge, the mean observed gain, the correlation of initial and post-process knowledge, and the correlation of observed gain with initial knowledge. All five empirical features use participants with knowledge scores at both waves. A simulated process passes the filter if each feature falls within the corresponding range across polls; it need not resemble any single poll jointly. Of 4,000 processes with proportional learning, 30% are plausible; with learning unrelated to initial knowledge, 6%; with catch-up learning, 8 of 4,000 processes. Proportional learning passes more often in this grid, but these rates are conditional on the chosen model family and parameter distribution. They do not establish which learning process generated the polls. The gain–initial-knowledge correlation is negative in every poll (-.66 to -.35). Ceilings can produce this pattern, but so can baseline measurement error, because the baseline score is subtracted in observed gain.

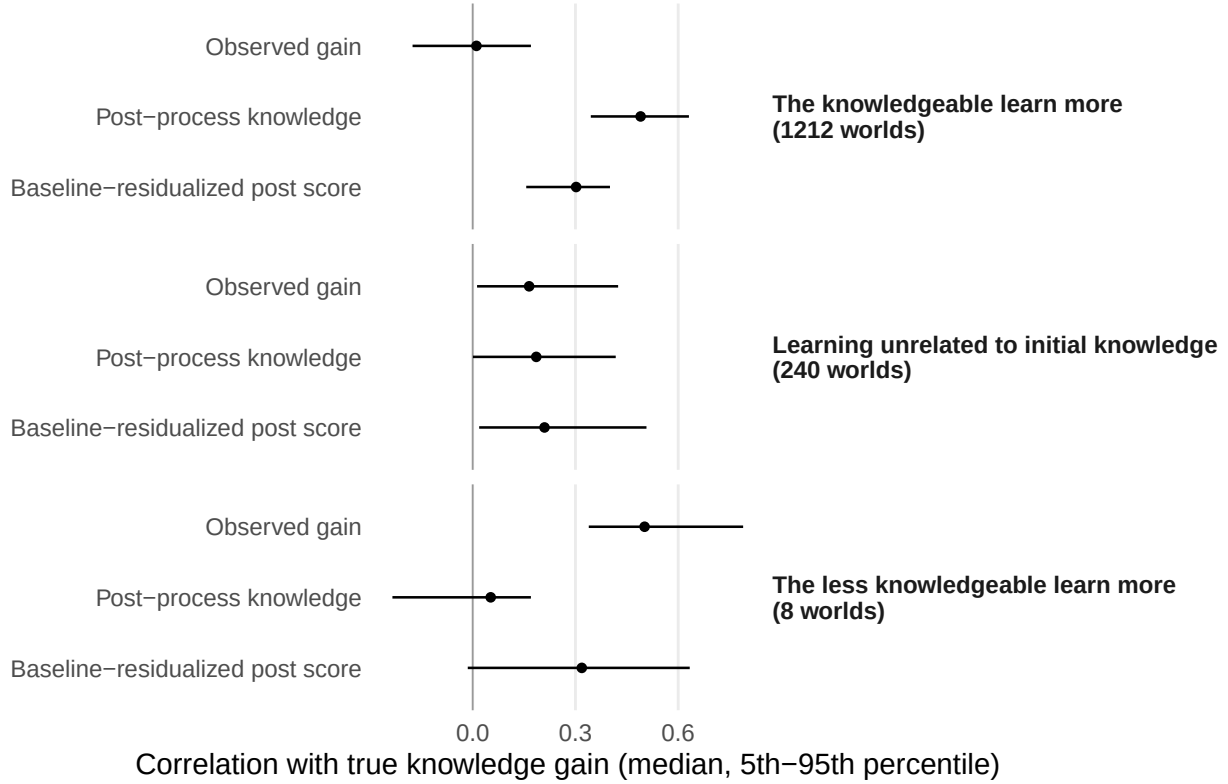


Figure 1: Correlation of each proxy with true knowledge gain across simulated processes whose observable features fall within the marginal ranges across Deliberative Polls. Points are medians and lines span the 5th to 95th percentiles across processes; they describe variation across plausible worlds, not sampling uncertainty. The baseline-residualized score is the residual from regressing post-process on initial knowledge. Every score is correlated with the same, unresidualized true gain.

Figure 1 shows the results. When the knowledgeable learn more, observed gain’s median correlation with true gain is .01, negative in 46% of plausible processes; post-process knowledge’s is .49, never negative, and higher than observed gain’s in 100% of them. The baseline-residualized post score has median correlation .30 with true gain. The corresponding partial correlation is .34, but answers a different question: it also removes the variation in true gain associated with observed baseline knowledge. When learning is unrelated to initial knowledge, the three scores have median correlations .16, .19 and .21; observed gain performs best in the 8 retained catch-up processes. This small selected subset does not characterize catch-up learning generally.

Asking more items helps observed gain only slightly within this simulation grid. Among retained processes with proportional learning, its median correlation is .00 with 5 items and .03 with 15, while post-process knowledge’s is .45 and .55. These are comparisons across separately selected processes, not an experiment holding the process and item difficulty distribution fixed. They do not establish that a longer or better targeted test cannot measure learning well.

7 Does the Choice Matter?

Consider a question the literature on deliberation asks often: who learns? In the same 21 polls, we regress each learning proxy on whether education is strictly above the poll’s participant median,

poll by poll, and pool the estimates in a Bayesian random-effects meta-analysis (Röver 2020). With observed gain, the above-median education difference is uncertain (.017 [-.005, .041]), with a negative estimate in 6 of the polls. With post-process knowledge given initial knowledge, the above-median group scores higher (.066 [.049, .085]), with a negative estimate in 1 of 21 polls (Figure 2). Post-process knowledge alone, without initial knowledge, gives .116 [.098, .134], mixing what people learned with what they already knew. As the model implies, observed gain with initial knowledge held constant gives exactly the same estimate as post-process knowledge with initial knowledge held constant.

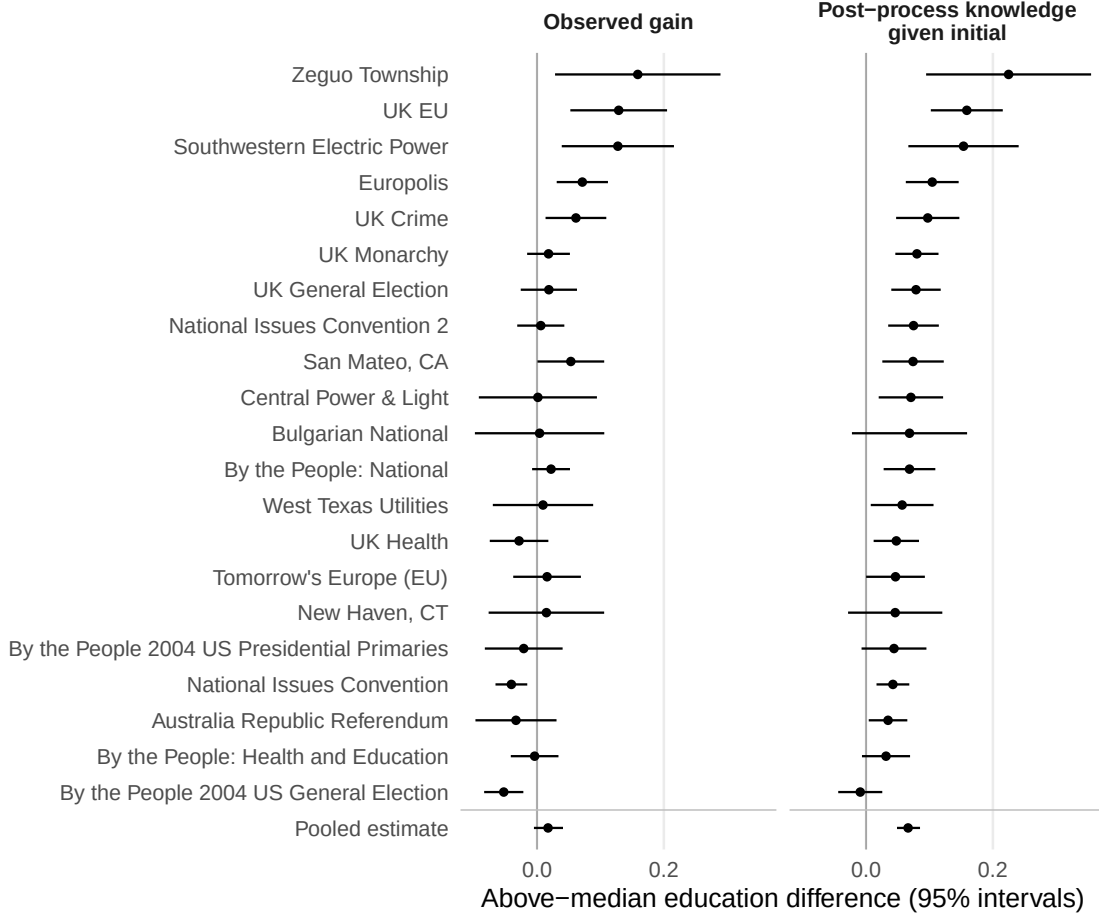


Figure 2: Difference in learning proxies for participants above versus at or below their poll's education median, by poll, with 95% intervals. Left: observed gain. Right: post-process knowledge holding initial knowledge constant. Standard errors are clustered by small group; the pooled estimates are from Bayesian random-effects meta-analyses.

The education indicator comes from dp-data's reviewed poll-specific ordered measures. Its fixed reference population is the unique historical-polardata participant sample with observed education, before exclusions for outcomes or group IDs; ties at the median remain in the lower group. We use the exported indicator without recomputing cutoffs. Education, paired knowledge and discussion-group membership must be observed for a participant to enter the regressions. The comparisons are associations with observed scores, not causal effects of education or direct measurements of true learning. Poll estimates use HC1 standard errors clustered by discussion group and normal 95%

intervals. The meta-analysis uses a half-normal prior with scale .1 on between-poll heterogeneity and an improper uniform prior on the pooled mean; its intervals are 95% posterior credible intervals.

Learning is often also used to explain attitude change: in Deliberative Polls, the participants who learn the most have been expected to account for much of the net change in opinion. We estimate these associations on all 129 attitude indices in the corrected data, regressing the change in each index on a learning proxy and the participant’s distance from the other complete-case members of the small group at baseline. Records with missing scores or group IDs and groups with fewer than two complete cases are excluded. The net direction and proxy standard deviation are computed within that same analysis sample. The share of indices on which the coefficient is positive is 50% for observed gain, 59% for post-process knowledge and 53% for post-process knowledge given initial knowledge; the shares significantly positive are 5%, 16% and 12%, respectively (Appendix Table B2). These summaries show proxy-specific associations; they do not establish that learning has no relationship with attitude change. Initial knowledge also has positive associations. The tests are unadjusted for multiple comparisons, indices within a poll are dependent, and the direction is estimated from the same sample. The significance shares therefore are descriptive rather than a calibrated test against chance.

8 Conclusion

The oversampling of easy items, easily reached ceilings, and the tendency of the most knowledgeable to learn the most can make observed gain a weak indicator of true gain. Post-process knowledge often tracks gain better in the simulated processes studied here, but neither its positive covariance nor its higher correlation makes it an estimate of the amount learned. Baseline adjustment answers a further, distinct question and can discard variation in learning that baseline knowledge predicts.

Estimating how much people learn requires a common measurement scale, a defined domain, and a model of response error. The poll comparisons establish associations with observed scores; they do not establish who learned more on an unobserved topic-wide scale or the causal effect of learning on attitudes. Appendix C formally assesses alternatives to post-process knowledge and reports a simulation comparison of a joint item-response estimator.

References

- Allison, Paul D. 1990. “Change Scores as Dependent Variables in Regression Analysis.” *Sociological Methodology* 20: 93–114. <https://doi.org/10.2307/271083>.
- Althaus, Scott L. 2003. *Collective Preferences in Democratic Politics: Opinion Surveys and the Will of the People*. Cambridge University Press. <https://doi.org/10.1017/cbo9780511610042>.
- Bartels, Larry M. 1996. “Uninformed Votes: Information Effects in Presidential Elections.” *American Journal of Political Science* 40 (1): 194–230. <https://doi.org/10.2307/2111700>.
- Bransford, John D., and Marcia K. Johnson. 1972. “Contextual Prerequisites for Understanding: Some Investigations of Comprehension and Recall.” *Journal of Verbal Learning and Verbal Behavior* 11 (6): 717–26. [https://doi.org/10.1016/s0022-5371\(72\)80006-9](https://doi.org/10.1016/s0022-5371(72)80006-9).
- Chalmers, R. Philip. 2012. “Mirt: A Multidimensional Item Response Theory Package for the r Environment.” *Journal of Statistical Software* 48 (6): 1–29. <https://doi.org/10.18637/jss.v048.i06>.

- Cooke, Nancy J., Robert S. Atlas, David M. Lane, and Robert C. Berger. 1993. "Role of High-Level Knowledge in Memory for Chess Positions." *The American Journal of Psychology* 106 (3): 321–51. <https://doi.org/10.2307/1423181>.
- Cor, M. Ken, and Gaurav Sood. 2016. "Guessing and Forgetting: A Latent Class Model for Measuring Learning." *Political Analysis* 24 (2): 226–42. <https://doi.org/10.1093/pan/mpw010>.
- Delli Carpini, Michael X., and Scott Keeter. 1996. *What Americans Know about Politics and Why It Matters*. Yale University Press.
- Dooling, D. James, and Roy Lachman. 1971. "Effects of Comprehension on Retention of Prose." *Journal of Experimental Psychology* 88 (2): 216–22. <https://doi.org/10.1037/h0030904>.
- Eckhardt, Beverly B., Mary R. Wood, and Robin Smith Jacobvitz. 1991. "Verbal Ability and Prior Knowledge: Contributions to Adults' Comprehension of Television." *Communication Research* 18 (5): 636–49. <https://doi.org/10.1177/009365091018005004>.
- Esary, J. D., F. Proschan, and D. W. Walkup. 1967. "Association of Random Variables, with Applications." *The Annals of Mathematical Statistics* 38 (5): 1466–74. <https://doi.org/10.1214/aoms/1177698701>.
- Gilens, Martin. 2001. "Political Ignorance and Collective Policy Preferences." *American Political Science Review* 95 (2): 379–96. <https://doi.org/10.1017/s0003055401002222>.
- Hambrick, David Z. 2003. "Why Are Some People More Knowledgeable Than Others? A Longitudinal Study of Knowledge Acquisition." *Memory & Cognition* 31 (6): 902–17. <https://doi.org/10.3758/bf03196444>.
- Harris, T. E. 1960. "A Lower Bound for the Critical Probability in a Certain Percolation Process." *Mathematical Proceedings of the Cambridge Philosophical Society* 56 (1): 13–20. <https://doi.org/10.1017/s0305004100034241>.
- Lord, Frederic M. 1967. "A Paradox in the Interpretation of Group Comparisons." *Psychological Bulletin* 68 (5): 304–5. <https://doi.org/10.1037/h0025105>.
- Luskin, Robert C., Gaurav Sood, James S. Fishkin, and Kyu S. Hahn. 2022. "Deliberative Distortions? Homogenization, Polarization, and Domination in Small Group Discussions." *British Journal of Political Science* 52 (3): 1205–25. <https://doi.org/10.1017/s0007123421000168>.
- Millsap, Roger E. 2010. "Testing Measurement Invariance Using Item Response Theory in Longitudinal Data: An Introduction." *Child Development Perspectives* 4 (1): 5–9. <https://doi.org/10.1111/j.1750-8606.2009.00109.x>.
- Montgomery, Jacob M., and Josh Cutler. 2013. "Computerized Adaptive Testing for Public Opinion Surveys." *Political Analysis* 21 (2): 172–92. <https://doi.org/10.1093/pan/mps060>.
- Qian, Jiahe. 2018. *Modeling Growth with Adaptive Longitudinal Large-Scale Assessments*. RR-18-34. Educational Testing Service. https://www.ets.org/research/policy_research_reports/publications/report/2018/jzng.html.
- Recht, Donna R., and Lauren Leslie. 1988. "Effect of Prior Knowledge on Good and Poor Readers' Memory of Text." *Journal of Educational Psychology* 80 (1): 16–20. <https://doi.org/10.1037/0022-0663.80.1.16>.
- Röver, Christian. 2020. "Bayesian Random-Effects Meta-Analysis Using the Bayesmeta r Package." *Journal of Statistical Software* 93 (6). <https://doi.org/10.18637/jss.v093.i06>.

Appendix

A Proofs

A.1 Result 1 and Condition 1a

The correlation has the sign of its covariance. By (9),

$$\begin{aligned}\rho_{\Delta x \Delta X} < 0 &\iff (\beta - 1)(b_2\beta - b_1)\sigma_{X_1}^2 + b_2\sigma_v^2 < 0 \\ &\iff (\beta - 1)(b_1 - b_2\beta)\sigma_{X_1}^2 > b_2\sigma_v^2.\end{aligned}$$

The last inequality is Condition 1a. Its left side is positive by (7) and (8), so the covariance can be negative when the learning disturbance has sufficiently small variance.

A.2 Result 2

By (10),

$$\sigma_{x_2 \Delta X} = (\beta - 1)b_2\beta\sigma_{X_1}^2 + b_2\sigma_v^2 > 0.$$

Both addends are positive by (6) and (7). The correlation is therefore positive too.

A.3 Result 3 and Conditions 3a–3d

First, post-process knowledge always has the larger covariance with true gain:

$$\sigma_{x_2 \Delta X} - \sigma_{\Delta x \Delta X} = (\beta - 1)b_1\sigma_{X_1}^2 > 0.$$

If the gain covariance is nonpositive, Result 2 gives the desired correlation ordering immediately. If it is positive, the ordering also follows whenever post-process knowledge has the smaller variance. Thus

$$\sigma_{x_2}^2 < \sigma_{\Delta x}^2 \implies \rho_{x_2 \Delta X} > \rho_{\Delta x \Delta X}.$$

Conditions 3a and 3b. Subtracting (12) from (11) gives

$$\begin{aligned}\sigma_{\Delta x}^2 - \sigma_{x_2}^2 &= [(b_2\beta - b_1)^2 - b_2^2\beta^2]\sigma_{X_1}^2 + \sigma_{u_1}^2 \\ &= (b_1 - 2b_2\beta)b_1\sigma_{X_1}^2 + \sigma_{u_1}^2.\end{aligned}$$

Positivity is Condition 3a. Condition 3b makes the first term nonnegative; the strictly positive measurement-error variance makes the sum positive.

Condition 3c. Condition 1a and Result 2 imply

$$\rho_{\Delta x \Delta X} < 0 < \rho_{x_2 \Delta X}.$$

Condition 3d. Since $\Delta x_i = x_{i2} - x_{i1}$,

$$\begin{aligned}\sigma_{\Delta x}^2 - \sigma_{x_2}^2 > 0 &\iff \sigma_{x_1}^2 - 2\sigma_{x_1 x_2} > 0 \\ &\iff \gamma_{21} = \frac{\sigma_{x_1 x_2}}{\sigma_{x_1}^2} < \frac{1}{2}.\end{aligned}$$

A.4 Guessing

Write $k_i = \kappa r_i$. The score and gain become

$$y_{i2} = (1 - k_i)x_{i2} + k_i, \quad \Delta y_i = (1 - k_i)\Delta x_i.$$

Independence of k_i from the knowledge scores and true gain gives

$$\begin{aligned} \text{Cov}(\Delta y_i, \Delta X_i) &= E(1 - k_i) \text{Cov}(\Delta x_i, \Delta X_i), \\ \text{Cov}(y_{i2}, \Delta X_i) &= E(1 - k_i) \text{Cov}(x_{i2}, \Delta X_i). \end{aligned}$$

The multiplier is nonnegative and is positive unless everyone guesses correctly with probability one. The strict signs of Results 1 and 2 survive whenever it is positive.

A.5 Propositions 1 and 2

Fix the questionnaire's item difficulties. Let $m(s)$ be the expected time-2 score at true knowledge s , averaging over item-response and guessing randomness that is independent of (X_{i1}, R_i) . Under the monotone item-response model, m is nondecreasing. Define

$$q(x, r) = m(x + h(x, r)).$$

Because true gain is determined by (X_{i1}, R_i) , the law of total covariance gives

$$\text{Cov}(x_{i2}, \Delta X_i) = \text{Cov}(q(X_{i1}, R_i), h(X_{i1}, R_i)).$$

Proposition 1. If h is nondecreasing in both arguments, so is q . The variables X_{i1} and R_i are independent and hence associated (Esary et al. 1967). Two nondecreasing functions of associated variables have nonnegative covariance. Therefore

$$\text{Cov}(x_{i2}, \Delta X_i) \geq 0.$$

Proposition 2. Conditional on $X_{i1} = x$, both $q(x, R_i)$ and $h(x, R_i)$ are nondecreasing functions of the scalar R_i . Their covariance is nonnegative (Harris 1960), giving

$$\text{Cov}(x_{i2}, \Delta X_i \mid X_{i1} = x) \geq 0.$$

This proof does not require h to be nondecreasing in its first argument.

The conditioning variable in Proposition 2 is true initial knowledge. The result does not guarantee a nonnegative partial correlation after adjusting for noisy observed initial knowledge: that partial correlation is negative in 1 of the 1,212 retained proportional-learning simulations.

B The polls and further results

Table B1: Observable features of the Deliberative Polls. The first five judge which simulated processes are plausible; γ_{21} is the slope of the regression of x_2 on x_1 (Condition 3d).

Poll	Mean x_1	Mean gain	SD x_1	$r(x_1, x_2)$	$r(\Delta x, x_1)$	γ_{21}
San Mateo, CA	.15	.17	.14	.11	-.46	.18
Europolis	.20	.17	.16	.21	-.46	.29
Central Power & Light	.29	.16	.19	.27	-.62	.26
Zeguo Township	.31	.11	.31	.30	-.50	.35
By the People: Health and Education	.35	.06	.21	.35	-.55	.36
Tomorrow's Europe (EU)	.37	.18	.19	.47	-.44	.52
New Haven, CT	.37	.23	.18	.39	-.35	.53
National Issues Convention 2	.37	.12	.17	.56	-.44	.58
By the People: National	.39	.03	.18	.61	-.45	.60
West Texas Utilities	.40	.29	.24	.39	-.56	.38
Bulgarian National	.43	.09	.25	.29	-.58	.30
Southwestern Electric Power	.44	.20	.23	.22	-.57	.25
National Issues Convention	.48	.10	.20	.62	-.59	.52
Australia Republic Referendum	.48	.28	.23	.59	-.66	.46
By the People: US Primaries, 2004	.51	.12	.28	.72	-.42	.69
UK EU	.52	.20	.29	.45	-.52	.45
UK Crime	.52	.09	.23	.61	-.36	.67
UK General Election	.54	.10	.21	.53	-.52	.51
UK Monarchy	.65	.15	.19	.56	-.42	.60
UK Health	.66	.08	.23	.56	-.42	.59
By the People: US Election, 2004	.75	.07	.24	.71	-.51	.63

Table B2: Learning and attitude change. Each attitude index's change is regressed on the proxy and the distance from the small group's mean; the proxy's coefficient is signed so that positive means moving with the sample's net change. Significance at the 5% level with standard errors clustered by small group.

Proxy	Indices	Positive	Sig. positive	Sig. negative
Observed gain	129	.50	.05	.05
Post-process knowledge	129	.59	.16	.06
Post-process knowledge given initial	129	.53	.12	.04
Initial knowledge	129	.59	.10	.09

C Alternatives to post-process knowledge

C.1 Target and comparison criteria

The positive covariance of post-process knowledge with true gain does not establish that it is the best predictor. Let $\Delta = X_2 - X_1$ be true gain on a fixed knowledge scale and let $T_t = x_t$ denote the observed score at wave t , suppressing person subscripts. We compare predictors by their Pearson correlation with the same Δ and, when predictions are expressed in its units, by mean squared error $R(f) = E\{(\Delta - f)^2\}$. Assume finite second moments and positive variances wherever a correlation or variance ratio is used. Rank correlation is a separate criterion; the Pearson results below do not establish optimality for ranks.

The formal comparisons distinguish population predictors, whose defining moments or conditional distributions are known, from estimates obtained by fitting a model. The former establish how much information could help. The latter require identification and can lose that advantage through estimation error or misspecification.

C.2 Combining both scores in the linear model

Let $L_2 = a + bT_2$ be the best affine predictor of Δ using the follow-up score and let $L_{12} = c + w_1T_1 + w_2T_2$ be the best affine predictor using both scores. Allowing the baseline coefficient to be zero nests L_2 in the second class, so $R(L_{12}) \leq R(L_2)$. Calibrating both predictors to Δ makes this an error comparison in common units.

To characterize the gain, define the baseline residual after linear prediction from follow-up knowledge:

$$Z = (T_1 - ET_1) - \frac{\text{Cov}(T_1, T_2)}{\text{Var}(T_2)}(T_2 - ET_2).$$

Then $EZ = 0$ and $\text{Cov}(Z, T_2) = 0$. Orthogonality gives

$$L_{12} = L_2 + \frac{\text{Cov}(\Delta, Z)}{\text{Var}(Z)}Z, \tag{22}$$

$$R(L_2) - R(L_{12}) = \frac{\text{Cov}(\Delta, Z)^2}{\text{Var}(Z)}. \tag{23}$$

For a direct proof, put $q = \text{Cov}(\Delta, Z)$ and expand the risk of $L_2 + kZ$. Since $\text{Cov}(L_2, Z) = 0$,

$$R(L_2 + kZ) = R(L_2) - 2kq + k^2 \text{Var}(Z).$$

Its minimizer is $k = q/\text{Var}(Z)$, proving both expressions. Baseline strictly improves linear prediction if and only if $q \neq 0$.

The corresponding correlation comparison is

$$\text{Corr}(L_{12}, \Delta)^2 - \text{Corr}(T_2, \Delta)^2 = \frac{\text{Cov}(\Delta, Z)^2}{\text{Var}(Z) \text{Var}(\Delta)} \geq 0. \tag{24}$$

To obtain it, use $\text{Cov}(L_{12}, \Delta) = \text{Var}(L_{12})$ and $\text{Var}(L_{12}) = \text{Var}(L_2) + q^2/\text{Var}(Z)$. Under the paper's linear-model assumptions, T_2 has positive covariance with Δ , so the comparison also orders the unsquared correlations.

Using the notation of the linear model, let $z = (T_1 - ET_1, T_2 - ET_2)^\top$, $\Sigma = \text{Var}(z)$, and $c_\Delta = \text{Cov}(z, \Delta)$. The same predictor can be written

$$L_{12} = E\Delta + c_\Delta^\top \Sigma^{-1} z, \quad (25)$$

where

$$c_\Delta = \begin{pmatrix} b_1(\beta - 1)\sigma_{X_1}^2 \\ b_2\{\beta(\beta - 1)\sigma_{X_1}^2 + \sigma_v^2\} \end{pmatrix}.$$

The positive measurement-error variances and zero covariance restrictions make Σ positive definite. Writing $D = \det(\Sigma) > 0$, substitution yields

$$w_1 = \frac{b_1\sigma_{X_1}^2\{(\beta - 1)\sigma_{u_2}^2 - b_2^2\sigma_v^2\}}{D}. \quad (26)$$

Thus the baseline weight is positive when $(\beta - 1)\sigma_{u_2}^2 > b_2^2\sigma_v^2$ and negative when the inequality reverses. When baseline knowledge predicts learning and follow-up measurement error is relatively large, baseline helps recover knowledge associated with greater learning. When variation in the learning disturbance is relatively large, subtracting baseline helps distinguish newly acquired knowledge from knowledge already possessed. Neither the fixed weight of -1 in observed gain nor the zero weight in the follow-up score is generally optimal. Residualizing follow-up knowledge on baseline imposes another particular pair of weights and has no general advantage over T_2 for predicting the same true gain.

Equation (25) is a population benchmark. The two observed scores alone do not identify its measurement slopes, error variances, and learning moments. Repeated indicators, external calibration, or explicit sensitivity assumptions must supply that information. No normality assumption is needed for the best affine predictor; interpreting it as the full conditional mean requires more structure.

C.3 Conditional means using item responses

Write $\mathcal{Y} = (\mathbf{Y}_1, \mathbf{Y}_2)$ for all item responses at both waves. The population predictor with minimum mean squared error is

$$m_{12} = E[\Delta \mid \mathcal{Y}]. \quad (27)$$

For any square-integrable predictor $f(\mathcal{Y})$,

$$R(f) = R(m_{12}) + E\{[f(\mathcal{Y}) - m_{12}]^2\}. \quad (28)$$

Indeed, expanding the square leaves a cross term whose expectation is zero because $E(\Delta - m_{12} \mid \mathcal{Y}) = 0$. The minimizer is therefore m_{12} , uniquely up to events of probability zero.

Let $m_2 = E[\Delta \mid \mathbf{Y}_2]$ use only follow-up item responses and let $m_T = E[\Delta \mid T_2]$ use only their aggregate score. These predictors obey

$$R(m_{12}) \leq R(m_2) \leq R(m_T) \leq R(L_2).$$

Each step enlarges the information or function class available for prediction. In particular,

$$R(m_2) - R(m_{12}) = E[(m_{12} - m_2)^2], \quad (29)$$

$$R(m_T) - R(m_2) = E[(m_2 - m_T)^2]. \quad (30)$$

Baseline adds value exactly when it changes expected gain after conditioning on all follow-up responses. Item patterns add value beyond the follow-up total exactly when they change expected gain at a given total. If the total retains all the relevant information in the follow-up pattern, the second improvement is zero.

The conditional mean also maximizes squared Pearson correlation with true gain. For any square-integrable, nonconstant score $s(\mathcal{Y})$,

$$\text{Cov}(s, \Delta) = \text{Cov}(s, m_{12}).$$

Cauchy–Schwarz and $\text{Cov}(m_{12}, \Delta) = \text{Var}(m_{12})$ give

$$\text{Corr}(s, \Delta)^2 \leq \frac{\text{Var}(m_{12})}{\text{Var}(\Delta)} = \text{Corr}(m_{12}, \Delta)^2. \quad (31)$$

The final equality presumes m_{12} is nonconstant. If it is constant, every such score has zero covariance with gain. Positive affine transformations of a nonconstant m_{12} attain the same positive correlation, while only m_{12} itself minimizes squared error on the specified gain scale.

Estimating the two knowledge levels from the same joint posterior and then subtracting is equivalent to integrating gain:

$$E[X_2 \mid \mathcal{Y}] - E[X_1 \mid \mathcal{Y}] = E[X_2 - X_1 \mid \mathcal{Y}].$$

Scoring each level from its own wave instead gives $E[X_2 \mid \mathbf{Y}_2] - E[X_1 \mid \mathbf{Y}_1]$, which generally differs from m_{12} and has no general advantage over the follow-up score. Transforming posterior mean abilities before differencing also generally differs from integrating gain when the transformation to knowledge is nonlinear.

These results concern the true joint distribution. To see why a fitted model need not inherit them, let $\widehat{m}_{12}$ and $\widehat{m}_2$ be trained on data independent of the scored respondent. Averaging over training and test data, (28) implies

$$\begin{aligned} R(\widehat{m}_{12}) - R(\widehat{m}_2) &= -E[(m_{12} - m_2)^2] \\ &\quad + E[(\widehat{m}_{12} - m_{12})^2] - E[(\widehat{m}_2 - m_2)^2]. \end{aligned}$$

The information benefit can be outweighed by additional estimation error or misspecification. The conditional-mean theorem therefore supplies a benchmark, not a guarantee for a particular fitted IRT model.

C.4 What the assumptions identify

Under the general model, the response function is only required to be monotone. Rescaling true knowledge and the item difficulties by the same positive constant leaves every response probability unchanged while rescaling true gain. Thus the observations do not identify the amount learned without a scale calibration. Under proportional learning, even an increasing power transformation preserves the rich-get-richer structure:

$$X_{i2} = X_{i1}(1 + R_i) \implies X_{i2}^p - X_{i1}^p = X_{i1}^p [(1 + R_i)^p - 1], \quad p > 0.$$

Transforming the response function accordingly leaves the response distribution unchanged. Different powers can change which of two people with different initial knowledge and learning shocks gained more. A monotone knowledge scale therefore need not identify rankings of individual gains either.

This limitation is especially consequential for the proportion of *all* relevant facts known. A short, deliberately easy questionnaire does not identify its relation to that universe. The gamma trait used in the simulations supplies a scale by assumption. Its gains are meaningful within those simulated worlds, but are not estimates of the fraction of all political knowledge acquired by poll participants.

C.5 Guessing-adjusted follow-up knowledge

Suppose the same probability $g < 1$ of a correct guess applies to every person and item. The usual correction to the observed follow-up proportion y_{i2} is

$$\tilde{x}_{i2} = \frac{y_{i2} - g}{1 - g}.$$

It changes the units and level, but is a positive affine transformation. Therefore it has exactly the same ranking and correlation with true gain as y_{i2} . In our simulations, each process has one common guessing probability, so this correction cannot improve the follow-up score's correlation. Nor does it remove the ceiling. The statement concerns ranking and correlation, not bias in the mean proportion of questionnaire items known.

Item-specific or person-specific guessing probabilities can change the ranking. If Y_{ij2} is item correctness and $g_{ij} < 1$ the probability of guessing correctly when the item is not known, the corrected contribution is

$$\tilde{K}_{ij2} = \frac{Y_{ij2} - g_{ij}}{1 - g_{ij}}.$$

With a correct answer certain when the item is known, this contribution has conditional expectation equal to the item's binary knowledge state. When the item is unknown its conditional variance is $g_{ij}/(1 - g_{ij})$, so removing guessing bias can amplify noise. Estimated guessing probabilities add uncertainty. Negative corrected contributions should not simply be truncated to zero, which would undo the unbiasedness argument. If guessing varies with initial knowledge or the learning shock, the propositions' independence assumptions also need reconsideration. The item transition model of Cor and Sood (2016) uses both waves to estimate learning about the questions asked under a no-forgetting restriction; it does not by itself identify each person's topic-wide learning.

C.6 A joint item-response model

A joint IRT model fits the two waves together and estimates change directly. Let Y_{ijt} indicate a correct response and θ_{it} denote knowledge on a common latent scale. For example,

$$\Pr(Y_{ijt} = 1 \mid \theta_{it}) = g_j + (1 - g_j) \text{logit}^{-1}[a_j(\theta_{it} - d_j)], \quad (32)$$

where $a_j > 0$ is discrimination, d_j difficulty, and g_j a lower asymptote. Common anchor items link the scale across waves; their response parameters must be invariant, or changes in item functioning must be modeled. A joint distribution for $(\theta_{i1}, \theta_{i2})$ lets both waves inform each person's change. Measurement invariance is a substantive requirement for interpreting latent change (Millsap 2010; Qian 2018). Responses to the same item across waves may remain dependent even after conditioning on ability, so local independence also needs checking.

Let $X_{it} = \psi(\theta_{it})$ express knowledge on the target scale. The fitted model supplies

$$\widehat{\Delta X}_i = \int \int [\psi(\theta_2) - \psi(\theta_1)] \hat{p}(\theta_1, \theta_2 \mid y_i) d\theta_1 d\theta_2.$$

This integrates gain itself. For nonlinear ψ , it differs from transforming two posterior mean abilities and subtracting them. A known ψ is an identifying assumption; IRT alone cannot recover the relation to the proportion of all relevant facts known. Uncertainty must account for the posterior dependence between the two abilities and, for full inference, uncertainty in the item and population parameters. Fitting each wave separately, standardizing both to mean zero, and subtracting the scores would erase average change. Imposing nonnegative individual change or proportional learning in the prior would build that restriction into the answer; the data should also be fitted with a model allowing forgetting and different relations between baseline knowledge and gain.

A latent-standard-deviation change is not a percentage of knowledge. For an interpretable amount, define a fixed reference set of items and weights $\omega_j \geq 0$ summing to one. Under the knowledge-versus-guessing interpretation of (32), define

$$K(\theta) = \sum_j \omega_j \text{logit}^{-1}[a_j(\theta - d_j)].$$

Then estimate $E\{K(\theta_{i2}) - K(\theta_{i1}) \mid \mathbf{Y}_{i1}, \mathbf{Y}_{i2}\}$ and its interval. This expresses change in the expected proportion of the reference items known, with guessing excluded. The reference items and weights stay fixed across waves. Unlike a raw ability difference, this quantity is unchanged by rescaling the IRT ability and item parameters together. Generalization to a larger topic requires a defensible item-sampling frame or external calibration; the tested items alone cannot represent everything there is to know. A generic IRT lower asymptote also need not be literal random guessing, so that interpretation requires its own evidence.

The model should use incorrect and explicit “don’t know” answers separately when the response process supports that distinction, while treating absent questionnaires as missing. A freely estimated guessing parameter for every person is unlikely to be well determined by a short battery; item-level guessing parameters, constrained by the answer options or external calibration, are a more defensible starting point. To study who learns, education and other predictors can enter the joint change model, with measurement invariance checked across groups. Regressing separately shrunken person scores as though they were observed learning would discard their uncertainty and can distort those associations.

C.7 A head-to-head comparison

High agreement between IRT and conventional *level* scores cannot establish that their estimates of *change* perform alike. The relevant test is recovery of true gain on independent data, including respondents whose initial scores are already at the ceiling.

We implement (27) with a joint model for baseline log knowledge and its change. The baseline has mean zero and variance one, fixing the scale, and its change has an estimated mean, variance, and covariance with baseline. The pair has a working bivariate normal distribution. Knowledge is $X = \exp(\theta)$, a positive trait on this normalized scale, not the proportion of all knowable facts. Item slopes and intercepts are estimated subject to equality across waves; guessing probabilities are supplied by the test design. Slopes are constrained to $[0.2, 5]$ and change variance to be at least .0025. We use marginal likelihood estimation (Chalmers 2012), expanding the integration range when the fitted change distribution extends beyond it, followed by Gaussian quadrature to integrate gain. These are empirical Bayes estimates: the fitted item and population parameters are treated as known when scoring a new person.

Each replication uses 300 training respondents and 1,000 independent test respondents answering the same items. Baseline log knowledge is standard normal. Proportional, additive, and catch-up learning use the same positive lognormal rate mechanism as the earlier simulations, with median .5 and log-scale standard deviation .5. The main scenarios have 5, 10, or 15 easy items; further scenarios broaden the difficulty range, vary guessing across items, or make the follow-up items easier without telling the fitted model. Unlike the earlier grid, this comparison does not select simulations for resemblance to the polls. All predictions target the same realized $X_{i2} - X_{i1}$ in a given replication. The fitted normal distribution of log change is an approximation even in the proportional-learning scenarios, rather than access to the true simulation law.

We compare six procedures. Three are the original scores: raw gain, raw follow-up knowledge, and guessing-adjusted follow-up knowledge. The other three estimate an amount in the specified trait units: the joint posterior mean gain; the posterior mean gain using only follow-up responses; and the difference between posterior mean knowledge levels scored from each wave separately. The latter two use the same fitted model as the joint estimate, so this comparison holds item calibration fixed and changes how the response information is used. The fitted model sees no true knowledge or true gain. Raw questionnaire proportions and the positive knowledge trait have different units, so we compare all six by correlation and assess squared error only for the three estimates expressed in the target units.

Of 80 attempted fits, 75 yield converged, numerically stable models. Tables C1 and C2 compare methods on those same successful replications, with failures included in each scenario’s fit count. The reported summaries are medians across training/test replications, not confidence intervals. The repository also records rank correlations, mean error, conditional interval coverage, and results for respondents who answered every baseline item correctly.

Table C1: Correlation with true gain in independent test respondents. Entries are median correlations across successful replications; Fits reports successful fits over all attempts. Learning is proportional except in the additive and catch-up rows. All scenarios use 10 items unless stated otherwise.

Scenario	Fits	Raw gain	Follow-up	Guess adjusted	Joint IRT
Easy, 5 items	10/10	-.06	.35	.35	.44
Easy, 10 items	10/10	-.09	.41	.41	.55
Easy, 15 items	10/10	-.12	.43	.43	.63
Broader difficulties	10/10	-.02	.49	.49	.68
Additive learning	10/10	.12	.23	.23	.07
Catch-up learning	5/10	.62	.18	.18	.53
Item-specific guessing	10/10	-.10	.41	.39	.55
Easier follow-up items	10/10	-.19	.33	.33	.52

Table C2: Root mean squared error for estimates of the amount learned. Lower is better. All three procedures use the same estimated model and predict gain in the same positive-trait units. Entries are medians over the successful replications in the correlation table. Raw score proportions are not in these units and are not assigned an RMSE.

Scenario	Joint gain	Follow-up only	Separate-wave difference
Easy, 5 items	1.72	1.65	2.18
Easy, 10 items	1.18	1.23	1.88
Easy, 15 items	1.17	1.24	1.96
Broader difficulties	1.09	1.19	2.04
Additive learning	0.76	0.61	1.28
Catch-up learning	1.78	2.05	1.76
Item-specific guessing	1.54	1.52	2.13
Easier follow-up items	2.16	2.27	2.86

With 10 easy items and proportional learning, the joint estimate has median correlation 0.55 with true gain, compared with 0.41 for the follow-up score and -0.09 for raw gain. Its median RMSE is 1.18, versus 1.23 using only follow-up responses; it has lower RMSE in 9 of the 10 paired replications. More items and a broader difficulty range improve recovery. With only five easy items, however, the joint estimate’s higher correlation does not translate into lower median squared error than the follow-up-only model.

The departures matter. Under additive learning, the fitted joint estimator recovers little of the variation in true gain. Under catch-up learning, raw gain has the larger correlation, and the fitting software rejects some replications because a follow-up item is answered correctly by every training respondent. Making the follow-up items easier produces positive bias of 0.93 trait units and reduces joint conditional interval coverage to 75.8 percent. These intervals condition on estimated parameters; they are not full confidence intervals. Item-specific guessing adjustment also performs slightly worse than the raw follow-up score in this design.

Thus the conditional mean is the optimal target under squared loss, but this fitted normal-change implementation is not uniformly best. The results support using both waves when the measurement and learning models are adequate. They also make flexible change distributions, item-invariance

checks, and harder questions substantive parts of estimation. Ten replications per scenario provide an initial comparison, not a precise estimate of performance differences or a selection of one model for the polls.

C.8 What to do at the ceiling

IRT cannot create information where the questions supply almost none. When a participant answers every easy item correctly, large ranges of high ability have almost the same likelihood. A precise estimate of additional learning may then come chiefly from the prior. Harder anchor items or an adaptive test can add information in this range (Montgomery and Cutler 2013); repeated easy items alone cannot resolve it.

Where the model is too weak to support a point estimate, bounds can be more informative. With deterministic threshold responses, no guessing, and externally calibrated item difficulties, each response pattern locates true knowledge in an interval $[L_{it}, U_{it}]$. Hence

$$\Delta X_i \in [L_{i2} - U_{i1}, U_{i2} - L_{i1}].$$

An independently justified no-forgetting assumption intersects this interval with $[0, \infty)$. A perfect score supplies a lower bound, with an upper bound only if one is known externally. With stochastic responses or guessing, these become probabilistic statements requiring an explicit response model, not deterministic bounds. Reporting wide uncertainty for a ceiling respondent is preferable to assigning that person either zero learning or a precise gain determined mainly by assumptions.